

Voluntary Report – Voluntary - Public Distribution

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Report Name: Japan's Proposed GHG Emission Values for Ethanol Derived from Additional Feedstocks

Country: Japan

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Report Highlights:

On April 17, 2026, Japan's Ministry of Economy, Trade and Industry (METI) proposed default greenhouse gas (GHG) emission values for three additional ethanol sources. Currently, METI has established GHG emission values for U.S. corn and Brazilian sugarcane ethanol. The three new additions are ethanol derived from Brazilian corn, Thai sugarcane, and Thai cassava. [FAS/Tokyo published the original report on May 28, 2026, but subsequently corrected an error in Table 1.]

Japan's Sophisticated Act:

[The Sophisticated Methods of Energy Supply Structure Act](#) (Sophisticated Act) encourages Japanese energy suppliers (e.g., oil refineries) to replace fossil fuels with renewable energy sources. Under the Sophisticated Act, the Ministry of Economy, Trade and Industry (METI) has introduced a series of notifications targeting transport liquid biofuel since 2011.

At the beginning of Japanese fiscal year (FY, April-March) 2023, METI enacted Notification 3.0, establishing liquid transportation biofuel standards for FY 2023 through FY 2027. In Notification 3.0 ([JA2023-0014](#)), METI committed to revising the greenhouse gas (GHG) emission value (i.e., carbon intensity value, or CI score) for the gasoline baseline amount (Notification 3.1). METI also promised to introduce default GHG emission values for additional ethanol derived from various feedstocks (Notification 3.2) ([JA2025-0035](#)).

In October 2025, METI enacted Notification 3.1 ([JA2025-0039](#)), which included the following updates:

- GHG Emission Value for Gasoline: Increased to 90.17 g-CO₂eq/MJ due to higher emissions from desulfurization operations in gasoline production.
- GHG Reduction Target: Raised to 60 percent.

For more details about the Sophisticated Act, see the [Japan Biofuels Annual](#) GAIN Report.

Forthcoming Notification 3.2:

Currently, Notification 3.1 only includes default GHG emission values for U.S. corn-based ethanol and Brazilian sugarcane-based ethanol, which Japanese oil refineries have relied on for imported ethanol. However, METI seeks to expand the ethanol sourcing options available to Japanese oil refineries. METI requested a contracted advisor to calculate the default life-cycle GHG emission values for ethanol derived from Brazilian corn, Thai sugarcane, and Thai cassava to expand potential supply options ([JA2025-0035](#)).

On April 17, 2026, during the [21st Decarbonized Fuel Policy Subcommittee](#), METI presented the results of these calculations. Table 1 shows the summary results of the life-cycle GHG scores for ethanol derived from various crops. METI will draft Notification 3.2 and provide a public comment period before finalizing it.

Brazilian Corn-Based Ethanol:

According to [METI](#), 25 percent of Brazil's ethanol production is currently made from corn, while the remaining 75 percent is derived from sugarcane. Two Brazilian ethanol producers use corn as a feedstock. METI's contracted advisor has received life-cycle analysis (LCA) data from one of these producers to then calculate the default life-cycle GHG emission value based on Brazil's *RenovaCalc* model. The contracted advisor adopted 5.69 g-CO₂eq/MJ amount for ethanol transportation from Brazil to Japan, as specified in the current notification. The advisor adopted the notification's global warming potential (GWP) scores, 25 and 298, for CH₄ and N₂O, respectively, instead of using the Brazilian GWP

scores. METI plans to require that Brazilian corn-ethanol be derived from the second crop of soybean production.

The Brazilian producer that provided the LCA data to the advisor procures biomass for their ethanol production process, resulting in a GHG emission value for the ethanol production energy consumption of only 1.56 g-CO₂eq/MJ. However, if this or other producers use natural gas to produce ethanol, the resulting GHG emission value for the energy consumption during ethanol production is 13.81 g-CO₂eq/MJ. As a result, the proposed total life-cycle GHG emission values for Brazilian second-crop corn-derived ethanol is 28.59 g-CO₂eq/MJ (using biomass during production) and 40.40 g-CO₂eq/MJ (using natural gas during production).

Thai Ethanol:

Two Thai sugarcane-based ethanol producers and one cassava-based ethanol producer provided LCA data, which METI's contracted advisor used to calculate life-cycle GHG emission values using the GREET 2024 model.

Unlike Brazilian sugarcane-based ethanol producers, Thai companies generally produce ethanol from molasses rather than from cane juice. Consequently, the sugarcane-to-ethanol conversion ratio in Thailand is lower than in Brazil. Although Thai sugarcane-ethanol producers are located closer to Japan (compared to Brazil's distance to Japan), while also utilizing the byproducts of sugar production; the proposed life-cycle GHG emission score for Thai sugarcane-ethanol is higher than that of Brazilian sugarcane-ethanol.

Currently, Thai cassava-ethanol producers use wood chips as an energy source to produce ethanol, resulting in a proposed total life-cycle GHG value of 29.90 g-CO₂eq/MJ. However, if a producer uses natural gas instead of wood chips, the proposed value increases to 59.25 g-CO₂eq/MJ.

Land Use Change Penalties:

The existing biofuel standard (i.e., Notification 3.1) includes penalties when croplands are converted after April 1, 2012, from non-agricultural lands. Pertaining to the United States, if corn is derived from croplands converted from forested land or grasslands after April 2012, the land use change (LUC) penalty scores of 151.4 g-CO₂eq/MJ and 44.8 g-CO₂eq/MJ are applied, respectively. If Brazilian sugarcane is derived from croplands converted from forested land, the LUC penalty is 248.7 g-CO₂eq/MJ. For all proposed new ethanol sources, if the feedstock is harvested from agricultural lands converted from non-agricultural lands after April 2012, the produced ethanol will not be qualified to use the proposed default values. Thus, it will be extremely difficult, if not impossible, to use any feedstock from land that has been converted after April 2012.

Table 1. Proposed Life-Cycle GHG Emission Values of On-Road Ethanol Derived from Various Sources in Notification 3.2
(Unit: g-CO₂eq/MJ)

		U.S. Corn	Brazilian 2 nd Crop Corn		Brazilian Sugarcane	Thai Sugarcane	Thai Cassava	
			NG	Biomass			NG	Wood Chip
Status		Existing	Proposed	Proposed	Existing	Proposed	Proposed	Proposed
Total Value		36.86	40.40	28.15	28.59	28.26	59.25	29.90
Crop Cultivation	Chemical Input	6.15	4.53		3.59	0.03	0	
	Fertilizer	8.74	10.60		10.02	11.30	17.96	
	Machinery	2.61	1.15		0.26	2.48	1.71	
	Burning	--	--		6.01	2.56	--	
	Subtotal	17.50	16.28		19.88	16.37	19.67	
Crop Transportation Subtotal		1.01	0.91		-- *	4.72	1.18	
Ethanol Production	Energy Consumption	10.33	13.81	1.56	0.41	3.08	33.62	4.28
	Chemical Production	1.07	0.23		0.55	1.64	2.32	
	Subtotal	11.40	14.04	1.79	0.96	4.72	35.94	6.60
Ethanol Transportation	Domestic	2.76	3.48		2.06	0.94		
	Shipping to Japan	4.20	5.69			1.50		
	Subtotal	6.96	9.17		7.75	2.44		
Land use change penalty for croplands converted from non-ag lands after 2012		+151.4 (from forestlands) +44.8 (from grasslands)	Not qualified to adopt default values		+248.7 from forestlands	Not qualified to adopt default values		
Based Model		GREET 2022	RenovaCalc			GREET 2024		

* The value is included in “crop cultivation.”

NG: natural gas

Note: In [Notification 3.1](#), the baseline gasoline GHG emission value is 90.17 g-CO₂eq/MJ and the reduction target is 60 percent.

Due to rounding, totals may not equal the sum of individual values.

Biomass Concerns:

Japan imports biomass, including wood chips and pellets, to generate electric power under METI's feed-in tariff (FIT) and feed-in premium (FIP) programs. METI has established strict environmental standards to ensure the biomass comes from sustainable sources. Currently, METI only permits the use of three third-party certifications—GGL, SBP, and PEFC—for woody biomass. METI has also calculated the [default GHG values](#) for biomass. Unfortunately, the LCA calculation methodology for Brazilian corn-ethanol and Thai cassava-ethanol is opaque, making it unclear if the Brazilian and Thai producers utilize the same rigorous requirements for their biomass that METI mandates for Japanese FIT/FIP users.

National GHG Emissions and Removals:

An important note is that METI's calculated GHG emission values are recorded in the producing countries and are not included in [Japanese national GHG emissions and removals](#). The GHG emissions generated by biofuels are not reported or accounted for in the energy sector of the consuming country as stipulated under the Intergovernmental Panel on Climate Change (IPCC) rules. See [Japan's Biofuels Annual](#) for further information about Japan's national GHG emissions.

Attachments:

No Attachments.